

# Zhenqiao Song

CMU – Pittsburgh, Pennsylvania, US

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## Education

### Carnegie Mellon University

Pittsburgh, US

Jan. 2024-May. 2027

- Ph.D. in Language Technologies Institute. Advisor: Prof. Lei Li
- Thesis: Foundation Models for Functional *De Novo* Protein Design
- Committee: Lei Li (Chair), Jian Ma (CMU), Olexandr Isayev (CMU), Yisong Yue (Caltech)

### University of California, Santa Barbara

Santa Barbara, US

Sep. 2022-Dec. 2023

- Ph.D. in Computer Science (transfer to CMU). Advisor: Prof. Lei Li

### Fudan University

Shanghai, China

Sep. 2017-Jun. 2020

- M.S. in Computer Science. Advisor: Prof. Xiaoqing Zheng
- Shanghai Outstanding Graduate Student

### Xiamen University

Xiamen, China

Sep. 2013-Jun. 2017

- B.S. in Computer Science. Advisor: Prof. Jinsong Su

## Research Interests

My primary research goal is to build **controllable AI systems to accelerate therapeutic discovery and advance our understanding of life sciences**. My research highlights the following perspectives: (1) **AI for Science**: Generative modeling for functional protein design and biomolecular surface learning. (2) **Generative AI & Foundation Models**: Genomic foundation models, and multi-modal modeling bridging natural language and life sciences. (3) **Natural Language Processing**: Multilingual NLP, machine translation, and text generation.

## Employment

### Google Deepmind

New York City, US

May. 2025-Aug. 2025

*Student researcher working with Ankur Parikh and David Belanger, Protein Function Team*

–Research Project on Enzyme Design

### NEC Laboratories America

Princeton, US

Jun. 2024-Dec. 2024

*Research intern working with Martin Renqiang Min, Machine Learning Department*

–Research Project on Protein Complex Design

### Broad Institute of Harvard and MIT

Boston, US

Jun. 2023-Dec. 2023

*Visting scientist intern working with Wengong Jin, Eric and Wendy Schmidt Center*

–Research Project on Protein Design based on Protein Surface

### ByteDance AI Lab

Shanghai, China

Jul. 2020- Aug. 2022

*Research scientist, working with Dr. Hao Zhou and Dr. Lei Li, MLNLC team.*

–Research Work on Deep Generative Models for Natural Language

–Industrial Applications on Natural Language Generation

- Multilingual Advertising Title Generation

- Implement a multilingual Ad title generation model incorporating eight languages, including English, Chinese, Japanese, Vietnamese, Russian and so on. It is used in an online advertising system of ByteDance, and the profit arises from \$4,000/day to \$200,000/day.

- Advertisement Classification System

- Implement an advertisement classification model incorporating over 200 categories. It achieves over 90% online average accuracy on all categories and the profit is over 4 million dollars one year.

## –Research Work on Deep Generative Models for advertising bidding

- Developing the bidword generation model in the Sponsored Search Auction system of ByteDance and achieves over \$300 million every year.

**Publications** (\* for Equal Contribution, # for Project Lead)

- [1] Wenduo Cheng\*, **Zhenqiao Song\***, Yang Zhang\*, Shike Wang, Danqing Wang, Muyu Yang, Lei Li, and Jian Ma. “DNALONGBENCH: A Benchmark Suite for Long-Range DNA Prediction Tasks”. In: *Nature communications*. Nature Publishing Group UK London. **Featured for the Nature Communications Editors’ Highlights collection**, 2025.
- [2] **Zhenqiao Song**, Tianxiao Li, Lei Li, and Martin Renqiang Min. “PPDiff: Diffusing in Hybrid Sequence-Structure Space for Protein-Protein Complex Design”. In: *Forty-second International Conference on Machine Learning (ICML)*. 2025.
- [3] Qihao Duan, Bingding Huang, **Zhenqiao Song**, Irina Lehmann, Lei Gu Gu, Rol Eils, and Benjamin Wild. “JanusDNA: A Powerful Bi-directional Hybrid DNA Foundation Model”. In: *Thirty-ninth Conference on Neural Information Processing Systems (NeurIPS)*. 2025.
- [4] **Zhenqiao Song**, Yunlong Zhao, Wengong Jin, Wenxian Shi, Yang Yang, and Lei Li. “Generative Enzyme Design Guided by Functionally Important Sites and Small-Molecule Substrates”. In: *Forty-first International Conference on Machine Learning (ICML)*. 2024.
- [5] **Zhenqiao Song**, Tinglin Huang, Lei Li, and Wengong Jin. “SurfPro: Functional Protein Design Based on Continuous Surface”. In: *Forty-first International Conference on Machine Learning (ICML)*. 2024.
- [6] Tinglin Huang, **Zhenqiao Song**, Rex Ying, and Wengong Jin. “Protein-Nucleic Acid Complex Modeling with Frame Averaging Transformer”. In: *Thirty-eighth Conference on Neural Information Processing Systems (NeurIPS)*. 2024.
- [7] Kexun Zhang, Yee Man Choi, **Zhenqiao Song**, Taiqi He, William Yang Wang, and Lei Li. “Hire a Linguist!: Learning Endangered Languages with In-Context Linguistic Descriptions”. In: *Proceedings of the 62th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2024.
- [8] **Zhenqiao Song** and Lei Li. “Importance Weighted Variational Bayes for Protein Sequence Design”. In: *Proceedings of the 2023 International Conference on Machine Learning (ICML)*. 2023.
- [9] Wenda Xu, Danqing Wang, Liangming Pan, **Zhenqiao Song**, Markus Freitag, William Yang Wang, and Lei Li. “Instructscore: Towards explainable text generation evaluation with automatic feedback”. In: *Proceedings of the 2023 EMNLP (EMNLP)*. 2023.
- [10] **Zhenqiao Song**, Hao Zhou, Lihua Qian, Jingjing Xu, Shanbo Cheng, Mingxuan Wang, and Lei Li. “switch-GLAT: Multilingual Parallel Machine Translation Via Code-Switch Decoder”. In: *International Conference on Learning Representations (ICLR)*. 2022.
- [11] Yiran Chen, **Zhenqiao Song**#, Xianze Wu, Danqing Wang, Jingjing Xu, Jiaze Chen, Hao Zhou, and Lei Li. “MTG: A Benchmarking Suite for Multilingual Text Generation”. In: *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL Findings)*. 2022.
- [12] **Zhenqiao Song**, Jiaze Chen, Hao Zhou, and Lei Li. “Triangular Bidword Generation for Sponsored Search Auction”. In: *Proceedings of the 14th ACM International Conference on Web Search and Data Mining (WSDM)*. 2021, pp. 707–715.
- [13] **Zhenqiao Song**, Xiaoqing Zheng, and Xuanjing Huang. “Jointly learning bilingual word embeddings and alignments”. In: *Machine Translation*. Springer, 2021, pp. 1–19.

- [14] Jingbing Wang, Zhuolin Hao, Minping Zhou, Jiaze Chen, Hao Zhou, **Zhenqiao Song**, Jinghao Wang, Jiandong Yang, and Shiguang Ni. “Generating personalized titles incorporating advertisement profile”. In: *Database Systems for Advanced Applications: 26th International Conference, DASFAA 2021, Taipei, Taiwan, April 11–14, 2021, Proceedings, Part III* 26. Springer. 2021, pp. 529–540.
- [15] **Zhenqiao Song**, Xiaoqing Zheng, Lu Liu, Mu Xu, and Xuan-Jing Huang. “Generating responses with a specific emotion in dialog”. In: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2019, pp. 3685–3695.
- [16] Jinsong Su, **Zhenqiao Song**, Yaojie Lu, Mu Xu, Changxing Wu, and Yidong Chen. “Exploring Implicit Semantic Constraints for Bilingual Word Embeddings”. In: *Neural Processing Letters*. Vol. 48. 2. Springer, 2018, pp. 1073–1088.

## Preprint

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- [17] **Zhenqiao Song\***, Huichong Liu\*, Yunlong Zhao\*, Yang Yang, and Lei Li. “Ligand-guided Sequence-structure Co-design of *De Novo* Functional Enzymes”. In: *bioRxiv* (2026), pp. 2026–03.
- [18] **Zhenqiao Song**, Ramith Hettiarachchi, Chuan Li, Jianwen Xie, and Lei Li. “Natural Language Guided Ligand-Binding Protein Design”. In: <https://arxiv.org/abs/2506.09332> (2025).
- [19] **Zhenqiao Song**, Yunlong Zhao, Wenxian Shi, Yang Yang, and Lei Li. “Functional Geometry Guided Protein Sequence and Backbone Structure Co-Design”. In: *arXiv preprint arXiv:2310.04343* (2023).
- [20] **Zhenqiao Song**, Yunlong Zhao, Yufei Song, Wenxian Shi, Yang Yang, and Lei Li. “Joint Design of Protein Sequence and Structure based on Motifs”. In: *arXiv preprint arXiv:2310.02546* (2023).
- [21] Lu Liu, **Zhenqiao Song**, and Xiaoqing Zheng. “Improving Coreference Resolution by Leveraging Entity-Centric Features with Graph Neural Networks and Second-order Inference”. In: *arXiv preprint arXiv:2009.04639* (2020).

## Invited Talk

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- **Talk Title: Generative AI for Functional Protein Design**
  - Sep. 2025: AstraZeneca
  - Jun. 2025: Google Deepmind Protein Design/Function Team
  - Dec. 2025: ByteDance drug discovery team
  - Apr. 2025: Tsinghua University Air Institute GenSI open day
  - Aug. 2024: Tsinghua University Air Institute
  - Jul. 2024: Fudan University NLP Group
- **Talk Title: Importance Weighted Expectation-Maximization for Protein Sequence Design**
  - Oct. 2023: Jiangmen
  - May 2023: BytaDance Research

## Honors and Awards

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- **Lambda Lab Computing Grants. \$25,000.** 2026.
- **EECS Rising Stars 2025.** (hosted by MIT)
- **NEC Faculty Research Award. \$75,000.** 2024. PI: Lei Li. Contributed to ideation and grant writing.
- **China National Scholarship.** 2020
- **Shanghai Outstanding Graduate Student Award.** 2020

## Academic Services

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### ○ Organization

- **Organizer** of the Second Generative AI and Biology workshop (GenBio) in ICML 2025
- **Organizer** of the First Generative AI and Biology workshop (GenBio) in NeurIPS 2023
- **Organizer** of the 2022 SoCal (Southern California) NLP Symposium

### ○ Journal Reviewer

- Nature Communications
- Nature Machine Intelligence
- Medicine
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- IEEE Transactions on Computational Biology and Bioinformatics (TCBB)
- Transactions on Machine Learning Research (TMLR)

### ○ Conference Reviewer

- ICML 2023-2026, NeurIPS 2023-2025, ICLR 2023-2026, AISTATS 2025-2026, COLM 2024-2025, ACL 2020-2024, EMNLP 2020-2024, IJCAI 2023-2024, NLPCC 2022-2023

## Students Advising

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### ○ Graduate Students

- Charles Novak (2025.6 – Now, CMU bachelor -> CMU master)
- Jacky Chen (2024.9-2025.1, University of Pittsburgh PhD)
- Ramith Hettiarachchi (2024.9-2025.1, CMU PhD)
- Xiwei Cheng (2024.6-2025.4, UCSD master -> Northeastern University PhD)
- Yujia Gao (2024.1-2024.5, CMU master)

### ○ Undergraduate Students

- Jingyu Zhu (2024.6-2024.9, Peking University)
- Yufei Song (2023.1-2023.12, UCSB -> UCLA master)

### ○ Interns at ByteDance AI Lab

- Lu Liu (2021.6 - 2021.9, Fudan University Master -> ByteDance NLP researcher)
- Yiran Chen (2021.6 - 2021.9, Fudan University Master -> startup NLP researcher)

## Teaching

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### ○ Teaching Assistant at CMU

- CMU 11777, Multimodal Machine Learning at Spring 2026.
- CMU 11781/02741, Generative AI for Biomedicine at Fall 2025

### ○ Teaching Assistant at UCSB

- CS 190I Deep Learning at Winter 2023
- CS 165B Machine Learning at Spring 2023

### ○ Teaching Assistant at Fudan University

- Data Structures and Algorithms at Fall 2017
- Linear Algebra at Spring 2018